DIP2C-related Developmental Disorder With Speech Delay — Comprehensive Disease Report

MONDO ID: MONDO:0700356 · **Gene:** *DIP2C* (Disco-interacting protein 2 homolog C) · **Locus:** 10p15.3 · **OMIM gene:** 611380 · **HGNC:** 29150 · **Category:** Mendelian (autosomal dominant)

Summary

DIP2C-related Developmental Disorder With Speech Delay is an ultra-rare, autosomal-dominant neurodevelopmental disorder caused by heterozygous loss-of-function (haploinsufficiency) of *DIP2C* at chromosome 10p15.3. Its defining and most penetrant feature is a **developmental delay that preferentially affects expressive language and speech articulation**, frequently accompanied by variable global developmental delay, hypotonia, cognitive/behavioral differences, minor craniofacial dysmorphism, and — in a minority — seizures and congenital heart defects. The disorder was delineated as a discrete entity by Ha et al. (2024), who reported 23 individuals with heterozygous *DIP2C* variants (PMID: 38421105). *DIP2C* is also one of the two critical-region genes (alongside *ZMYND11*) of the overlapping **10p15.3 microdeletion syndrome** (PMID: 22847950).

The molecular mechanism is dosage sensitivity of a conserved lipid-metabolic enzyme. DIP2C belongs to the DISCO-interacting protein 2 (DIP2) family, which carries a DMAP1-binding domain, acyl-CoA synthetase domain, and AMP-binding (fatty-acyl-AMP-ligase, FAAL-like) sites. Across yeast, fly, worm, zebrafish, and mouse models, DIP2 proteins act as homeostatic regulators of a chemically distinct subset of **diacylglycerols (DAG)**, redirecting DAG flux into triacylglycerol, balancing *de novo* phospholipid synthesis, and controlling a DAG→PKC signaling axis. Loss of this function perturbs **axon bifurcation/guidance, axon maintenance/regeneration, and myelination**, providing a plausible causal chain from lipid dysregulation to disrupted corticolinguistic circuit maturation. gnomAD constraint metrics (pLI ≈ 1.0, LOEUF = 0.149, observed/expected LoF = 0.079) place *DIP2C* among the most loss-of-function-intolerant genes in the genome, quantitatively corroborating the haploinsufficiency model.

Clinically, the disorder is diagnosed molecularly — via chromosomal microarray (for 10p15.3 deletions) or exome/genome sequencing (for intragenic variants) — as there is no specific biochemical biomarker or imaging signature. Inheritance is autosomal dominant with both *de novo* and inherited variants (frequently from mildly or subclinically affected parents), with **incomplete penetrance and variable expressivity**. The core phenotype is **non-progressive and compatible with a normal lifespan**; morbidity derives from developmental disability and, in a subset, from cardiac defects or seizures. There is **no disease-specific or gene-targeted therapy** (0 registered clinical trials); management is entirely supportive, centered on early speech-language therapy, developmental support, and cardiac/seizure surveillance where indicated.

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1. Disease Information

Overview. DIP2C-related Developmental Disorder With Speech Delay is a Mendelian neurodevelopmental disorder in which heterozygous loss-of-function of *DIP2C* produces developmental delay with a characteristic predilection for expressive speech and language. It exists on a phenotypic continuum with the contiguous-gene **10p15.3 microdeletion syndrome**, in which *DIP2C* is co-deleted with the neighbouring *ZMYND11* gene.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0700356
Gene (OMIM)	*DIP2C* 611380
HGNC	29150
Ensembl	ENSG00000151240
Entrez Gene (human)	22982
Cytogenetic locus	10p15.3 (GRCh37 chr10:320,130–735,683)
Related syndrome	10p15.3 microdeletion syndrome (co-gene *ZMYND11*, OMIM 608668)

Synonyms / alternative names. DIP2C-related developmental disorder; DIP2C haploinsufficiency; expressive speech delay associated with DIP2C variants; (overlapping) 10p15.3 microdeletion syndrome; Disco-interacting protein 2 homolog C-related disorder.

Nature of information. The evidence base is derived from **aggregated individual case reports and small cohorts** (not EHR-scale datasets). The anchor dataset is a 23-individual cohort with intragenic variants (PMID: 38421105), supplemented by ~19 individuals with 10p15.3 deletions (PMID: 22847950) and scattered prenatal/postnatal case reports.

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2. Etiology

Primary cause — genetic. The disorder is caused by **heterozygous loss-of-function of *DIP2C*** through either intragenic variants (nonsense, frameshift, splice, and some missense) or contiguous 10p15.3 microdeletions encompassing *DIP2C*. The mechanism is **haploinsufficiency** — a single functional allele is insufficient for normal neurodevelopment. Ha et al. describe *"23 individuals with heterozygous DIP2C variants, all manifesting developmental delays that primarily affect expressive language and speech articulation"* (PMID: 38421105).

Genetic risk factors. In the defining cohort, the variant spectrum comprised *"Eight patients had de novo variants predicting loss-of-function in the DIP2C gene, two patients had de novo missense variants, three had paternally inherited loss of function variants and six had maternally inherited loss-of-function variants, while inheritance was unknown for four variants"* (PMID: 38421105). In 10p15.3 deletion cohorts, deletions were *de novo* in 7/9 tested (PMID: 22847950). A **family history** of a mildly affected parent is itself a risk indicator, reflecting incomplete penetrance (e.g., paternal transmission of a 10p15.3 deletion from a father with mild language impairment, PMID: 39696561).

Environmental risk / protective factors. No environmental risk or protective factors are established for this monogenic disorder; no modifier alleles or protective variants have been identified. Mendelian randomization in a mouse-centred study suggested a *positive causal relationship between linoleic acid and developmental disorders* (PMID: 41054236), but this is hypothesis-generating and not clinically actionable.

Gene–environment interactions. None demonstrated.

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3. Phenotypes

The phenotype is **neurodevelopmental and multisystem**, with a strong predilection for expressive speech. The table integrates data from the intragenic cohort (PMID: 38421105) and the 10p15.3 deletion cohort (PMID: 22847950). In the deletion series, *"Common features among these 12 individuals include: cognitive/behavioral/developmental differences (11/11), speech delay/language disorder (10/10), motor delay (10/10), craniofacial dysmorphism (9/12), hypotonia (7/11), brain anomalies (4/6) and seizures (3/7)"* (PMID: 22847950).

Phenotype	Type	Frequency	Suggested HPO term
Expressive language / speech-articulation delay	Behavioral / developmental	Core / near-universal (23/23 intragenic; 10/10 deletion)	HP:0002474 (Expressive language delay); HP:0000750
Global developmental delay	Developmental	Common	HP:0001263
Cognitive / intellectual / behavioral differences	Behavioral	11/11 (deletion cohort)	HP:0001249; HP:0000708
Motor delay	Developmental	10/10 (deletion cohort)	HP:0001270
Craniofacial dysmorphism (high anterior hairline/long forehead, broad nasal tip, ear anomalies)	Physical	9/12 (deletion); minor/inconsistent (intragenic)	HP:0001999
Hypotonia	Clinical sign	7/11 (deletion cohort)	HP:0001252
Brain anomalies (imaging)	Clinical sign	4/6 (deletion cohort)	HP:0012443
Seizures	Clinical sign	3/7 (deletion); focal infantile epilepsy reported	HP:0001250
Congenital heart defects (HCM, ASD, bicuspid aortic valve)	Physical	4/23 (intragenic cohort)	HP:0001627; HP:0001631; HP:0001647
Autistic-like traits / ASD	Behavioral	Reported subset	HP:0000729

Phenotype characteristics. Onset is **congenital/early-childhood**; severity is **variable (mild to moderate)**; progression is **stable/non-progressive**. Expressive speech delay is the most consistent and penetrant feature; the remaining features show incomplete penetrance and variable expressivity. The cardiac component is documented: *"Four patients had cardiac defects (hypertrophic cardiomyopathy, atrial septal defects, and bicuspid aortic valve)"* (PMID: 38421105).

Quality-of-life impact. The dominant QoL burden stems from **communication impairment** (expressive speech/language), affecting education, socialization, and independence; motor delay and behavioral/cognitive issues add functional burden. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist.

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4. Genetic / Molecular Information

Causal gene. *DIP2C* (Disco-interacting protein 2 homolog C), OMIM 611380, HGNC:29150, Ensembl ENSG00000151240, chromosome 10p15.3. In the contiguous deletion syndrome, *ZMYND11* (OMIM 608668) is co-deleted and contributes to the shared phenotype (PMID: 22847950, PMID: 34818214). DeScipio et al. established that *"Molecular mapping of the 19 individuals reported in this study has identified two genes, ZMYND11 (OMIM 608668) and DIP2C (OMIM 611380) ... mapping within 10p15.3 which are most commonly deleted"* (PMID: 22847950).

Pathogenic variants. - **Types:** loss-of-function predominates — nonsense, frameshift, canonical splice-site — plus *de novo* missense and whole-/partial-gene deletions. A specific *de novo* splice variant **NM_014974.3:c.1057+2T>G** (minigene-confirmed 80-bp exon-8 skip) was reported in focal infantile epilepsy (PMID: 34617658). - **Classification:** truncating variants in this highly constrained gene are generally pathogenic/likely pathogenic by ACMG criteria (PVS1-applicable for LoF); missense variants require case-by-case evaluation. - **Allele frequency:** pathogenic variants are ultra-rare/absent in population databases; *DIP2C* is strongly depleted of LoF variants in gnomAD. - **Origin:** germline (both *de novo* and inherited). - **Functional consequence: loss of function / haploinsufficiency.**

Constraint (gnomAD, GRCh37). pLI = 1.00 (0.99999999); LOEUF (oe_lof upper 90% CI) = **0.149**; observed/expected LoF = **0.079** (7 observed vs 88.1 expected PTVs); LoF Z = 8.01. The gene is also **missense-constrained** (missense Z = 4.67, oe_mis = 0.58). *DIP2C* sits in the most LoF-constrained decile of the genome — quantitative support for haploinsufficiency (Finding F005).

Protein architecture. DIP2 proteins *"contain a DNA methyltransferase-associated protein 1 (DMAP1) binding domain, Acyl-CoA synthetase domain and AMP-binding sites"* (PMID: 38421105). The AMP-synthetase/FAAL-like domains confer catalytic activity on fatty acids/acyl-CoA: *"the AMP-synthetase domains of DIP2 are essential for its function, which may involve exerting a catalytic activity that modifies fatty acids"* (PMID: 27908785).

Modifier genes. None established. In the deletion syndrome, co-deletion of *ZMYND11* (a chromatin reader / transcriptional regulator) likely modifies and augments the phenotype.

Epigenetic information. The DMAP1-binding domain links DIP2C to a chromatin-associated complex, but no disease-specific DNA-methylation signature (episignature) has been validated.

Chromosomal abnormalities. 10p15.3 submicroscopic microdeletions (ranging from intragenic to multi-megabase) involving *DIP2C* ± *ZMYND11*; also described in the context of ring chromosome 10 / monosomy 10 mosaicism (PMID: 40049822) and inv dup del(10p) (PMID: 31542096).

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5. Environmental Information

This is a **monogenic disorder with no established environmental, lifestyle, or infectious contribution.** No toxins, exposures, dietary, or behavioral factors are known to cause or trigger it. Infectious agents are not applicable.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Heterozygous LoF variant or 10p15.3 deletion** reduces *DIP2C* gene dosage → **leads to** ~50% reduction in functional DIP2C protein (haploinsufficiency; gnomAD pLI ≈ 1.0, LOEUF 0.149).
2. Reduced DIP2C (a FAAL-like acyl-CoA/AMP-binding enzyme) → **results in** impaired homeostatic regulation of a chemically distinct **diacylglycerol (DAG)** subset. DIP2 is *"a homeostatic regulator of a chemically distinct subset of DAGs using yeast, fly, and mouse models"* (PMID: 35766356).
3. Impaired DAG handling → **leads to** two coupled consequences:
4. **(3a)** reduced redirection of DAG flux into triacylglycerol — *"the fatty acyl-AMP ligase-like domains of DIP2 are essential for the redirection of the flux of DAG subspecies to storage lipid, triacylglycerols"* (PMID: 35766356) — risking toxic DAG/ER stress; and
5. **(3b)** dysregulation of a specific DAG pool required for **PKC activation**: *"Dip2, a conserved DAG regulator across Fungi and Animalia, has emerged as a modulator of PKC signalling"* (PMID: 40327034).

6. Loss of DIP2 lipid control → **results in** failure to counterbalance *de novo* **phospholipid (Kennedy-pathway) synthesis**: *"loss of function in dip-2 led to suppression of the axon regrowth phenotype observed in either cept-2 or ept-2 mutants, suggesting that DIP-2 acts to counterbalance phospholipid synthesis"* (PMID: 39974891).
7. Perturbed membrane-lipid/DAG/PKC signaling → **leads to** defective **axon bifurcation and guidance**: *"DISCO Interacting Protein 2 (DIP2) is required for precise axonal bifurcation in Drosophila mushroom body (MB) neurons by suppressing ectopic bifurcation and regulating the guidance of sister axons"* (PMID: 27908785), acting partly through the downstream phospholipase-D-superfamily effector **Glaikit** (PMID: 28465232).
8. *Branch (mouse cortex):* DIP2C deficiency → **results in** *"a significant upregulation of sphingolipid metabolism in the cortex, accompanied by elevated oligodendrogenesis and myelination"* (via increased serine palmitoyltransferase, the rate-limiting enzyme) (PMID: 41054236).
9. Aggregate disruption of axonal wiring and myelination during cortical development → **leads to** abnormal maturation of **corticolinguistic/cognitive circuits → expressive speech/language delay, developmental delay, cognitive/behavioral differences** (inferred mechanistic link bridging animal-model lipid/axon phenotypes and the human phenotype).
10. *Parallel branch (cardiac/craniofacial):* DIP2C deficiency during development → **results in** congenital heart defects and craniofacial malformations, directly demonstrated by *dip2ca* knockdown in zebrafish — *"developmental defects ... include decreased body length, craniofacial malformations, and heart defects, including pericardial edema, a linear heart tube, looping defects, and arrhythmia"*, rescued by wild-type *dip2ca* (PMID: 40915331).

Detail by category

- **Molecular pathways:** DAG homeostasis → DAG→**PKC** signaling; **Kennedy pathway** (de novo phospholipid synthesis) counterbalance; **sphingolipid biosynthesis** (serine palmitoyltransferase); glycerolipid/triacylglycerol storage flux.
- **Cellular processes:** axon bifurcation/guidance, axon maintenance/regeneration, oligodendrocyte differentiation/myelination, lipid storage, ER-stress avoidance.
- **Protein dysfunction:** haploinsufficiency / loss of function of a FAAL-like acyl-CoA/AMP-binding enzyme — reduced catalytic lipid-modifying activity rather than a toxic gain of function.
- **Metabolic changes:** altered diacylglycerol, triacylglycerol, phospholipid, acyl-CoA, and sphingolipid pools; abnormal lipid metabolism documented in homozygous mutant mice.

- **Subcellular localization:** DIP2 localizes to **mitochondria–vacuole contact sites** (PMID: 35766356).
- **Immune / tissue-damage mechanisms:** not a primary feature; IMPC heterozygous mice showed incidental immune readouts (increased monocytes, altered lymphoid organ size) of uncertain relevance.

Suggested GO terms: GO:0046339 (diacylglycerol metabolic process), GO:0006654 (phosphatidic acid biosynthetic process), GO:0030148 (sphingolipid biosynthetic process), GO:0042552 (myelination), GO:0007411 (axon guidance), GO:0007409 (axonogenesis). **Suggested CL terms:** CL:0000540 (neuron), CL:0000128 (oligodendrocyte), CL:0000679 (glutamatergic neuron).

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7. Anatomical Structures Affected

- **Organ level — primary: brain / central nervous system** (UBERON:0000955 brain; UBERON:0001950 neocortex). Secondary/variable: **heart** (UBERON:0000948) with congenital defects.
- **Body systems:** nervous system (primary); cardiovascular system (subset); craniofacial/skeletal (dysmorphism); gastrointestinal (atypical gastroparesis reported in one adult with 10p15.3 deletion, PMID: 39958070).
- **Tissue / cell level:** nervous tissue — cortical neurons and **oligodendrocytes** (myelination); affected lipid metabolism in neural cells.
- **Subcellular level (GO Cellular Component):** mitochondrion (GO:0005739), ER membrane (GO:0005789), mitochondria-associated/contact sites; cytoplasmic lipid storage.
- **Localization / lateralization:** CNS involvement is **bilateral/diffuse**; cardiac and craniofacial involvement are structure-specific.

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8. Temporal Development

- **Onset: congenital / early childhood**; developmental delays become apparent as speech and motor milestones are missed. Seizures, when present, can be infantile-onset (PMID: 34617658).

- **Onset pattern:** insidious/developmental.
- **Progression:** the core neurodevelopmental phenotype is **stable / non-progressive** — a static developmental difference rather than a neurodegenerative process. Children typically make developmental progress with intervention.
- **Disease course / duration: chronic, lifelong** developmental disability of variable severity; lifespan is generally normal absent severe cardiac disease.
- **Critical periods:** the **early-childhood window** for speech/language and motor development is the key period for therapeutic intervention.

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9. Inheritance and Population

- **Epidemiology: ultra-rare.** The literature comprises ~23 individuals with intragenic *DIP2C* variants plus scattered 10p15.3-deletion cases. True prevalence and incidence are **unknown**; no reliable per-100,000 estimate exists.
- **Inheritance pattern: autosomal dominant.** Both *de novo* and inherited (paternal and maternal) LoF variants occur.
- **Penetrance: incomplete** — mildly or subclinically affected transmitting parents are documented: *"Parental CMA analysis revealed that the 10p15.3 microdeletion was inherited from the father, who displayed mild language impairment"* (PMID: 39696561).
- **Expressivity: variable** (from mild isolated speech delay to multisystem involvement with cardiac defects/seizures).
- **Genetic anticipation / repeat expansion:** not applicable.
- **Germline mosaicism / founder effects / consanguinity:** no established role; not an AR disorder, so consanguinity and carrier-frequency metrics are not applicable.
- **Population demographics:** no ethnic predilection established; cases reported across multiple populations (European, Chinese Han, others). **Sex ratio:** no predilection established. **Age distribution:** diagnosed in childhood.

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10. Diagnostics

Diagnosis is molecular/genomic. There is **no specific biochemical biomarker or imaging signature.**

- **Chromosomal microarray (CMA):** first-line for detecting 10p15.3 microdeletions involving *DIP2C* ± *ZMYND11* — *"We describe 19 unrelated individuals with submicroscopic deletions involving 10p15.3 characterized by chromosomal microarray (CMA)"* (PMID: 22847950, also PMID: 39696561, PMID: 30950019).
- **Exome / genome sequencing (WES/WGS):** detects intragenic SNVs/indels (nonsense, frameshift, splice, missense), used to delineate the cohort and identify the epilepsy-associated splice variant (PMID: 38421105, PMID: 34617658, PMID: 38300321).
- **Prenatal testing:** CMA on amniocytes has detected 10p15.3 deletions prenatally (PMID: 39696561, PMID: 40049822, PMID: 31542096); karyotyping/FISH characterize structural rearrangements (ring 10, inv dup del(10p)).
- **Functional confirmation:** minigene splice assays confirmed the pathogenic splice effect of c.1057+2T>G (PMID: 34617658).
- **Adjunct evaluations:** brain MRI (variable anomalies), echocardiography (for cardiac defects), EEG (if seizures), developmental/speech assessment.

Clinical criteria / differential diagnosis. No formal consensus diagnostic criteria exist; diagnosis rests on a compatible neurodevelopmental phenotype plus a pathogenic *DIP2C* variant/deletion. Differential diagnosis includes other causes of childhood apraxia of speech / expressive language disorder and other contiguous-gene/neurodevelopmental syndromes; within 10p15.3 deletions, the overlapping contribution of *ZMYND11* must be considered.

Screening. No population newborn or carrier screening exists; cascade testing of relatives is appropriate once a familial variant is identified.

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11. Outcome / Prognosis

- **Survival / life expectancy:** generally **normal lifespan**; the core speech/developmental phenotype is non-life-limiting. Prognosis may be modified by severity of congenital heart disease in the minority affected.

- **Morbidity / disability:** chronic developmental disability dominated by **communication impairment**, with variable cognitive, motor, and behavioral contributions. Many children improve with early intervention.
- **Complications:** seizures (subset), congenital heart defects (subset, ~4/23 in the intragenic cohort), feeding difficulties, and rare gastrointestinal dysmotility (gastroparesis) in deletion cases.
- **Recovery potential:** the disorder is static; targeted therapies (speech, OT/PT) improve function but do not cure the underlying developmental difference.
- **Prognostic factors:** extent of the genomic lesion (intragenic variant vs large deletion co-involving *ZMYND11*), presence of structural brain anomalies, cardiac defects, and seizures. No validated molecular prognostic biomarker exists.

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12. Treatment

No disease-specific or gene-targeted therapy exists. A ClinicalTrials.gov search returned **0 studies** for "DIP2C" or "10p15.3 microdeletion." Management is **supportive and multidisciplinary**:

Intervention	Purpose	Suggested NCIT concept
Speech-language therapy	Address core expressive speech/ language delay	NCIT:C15376 (Speech Therapy)
Occupational / physical therapy	Motor delay, hypotonia, daily function	NCIT:C15277 (Physical Therapy); Occupational Therapy
Developmental / behavioral / educational support	Cognitive and behavioral needs, ASD traits	NCIT:C15312 (Supportive Care)
Antiseizure medication	For the seizure subset	NCIT:C264 (Anticonvulsant Agent)
Cardiac evaluation/management (echocardiography; surgery if indicated)	Congenital heart defects	NCIT:C49237 (Echocardiography)
Nutritional / GI management	Feeding difficulty, dysmotility	NCIT:C15431 (Nutritional Support)

- **Pharmacotherapy / pharmacogenomics:** none specific.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno):** none; all theoretical at present.
- **Treatment strategy:** individualized early-intervention program plus surveillance for cardiac and seizure complications.

13. Prevention

- **Primary prevention:** not applicable for a *de novo* genetic disorder beyond general preconception counseling.
- **Secondary prevention:** early developmental surveillance and prompt initiation of therapy to optimize outcomes.
- **Tertiary prevention:** cardiac and seizure surveillance to prevent complications.

- **Genetic counseling: key preventive measure.** Given autosomal-dominant inheritance with incomplete penetrance, counsel on up-to-50% recurrence risk when a parent carries the variant, the possibility of mildly affected carriers, and options including prenatal diagnosis and preimplantation genetic testing once a familial variant is known. Cascade testing of at-risk relatives is appropriate.
- **Immunization / public-health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

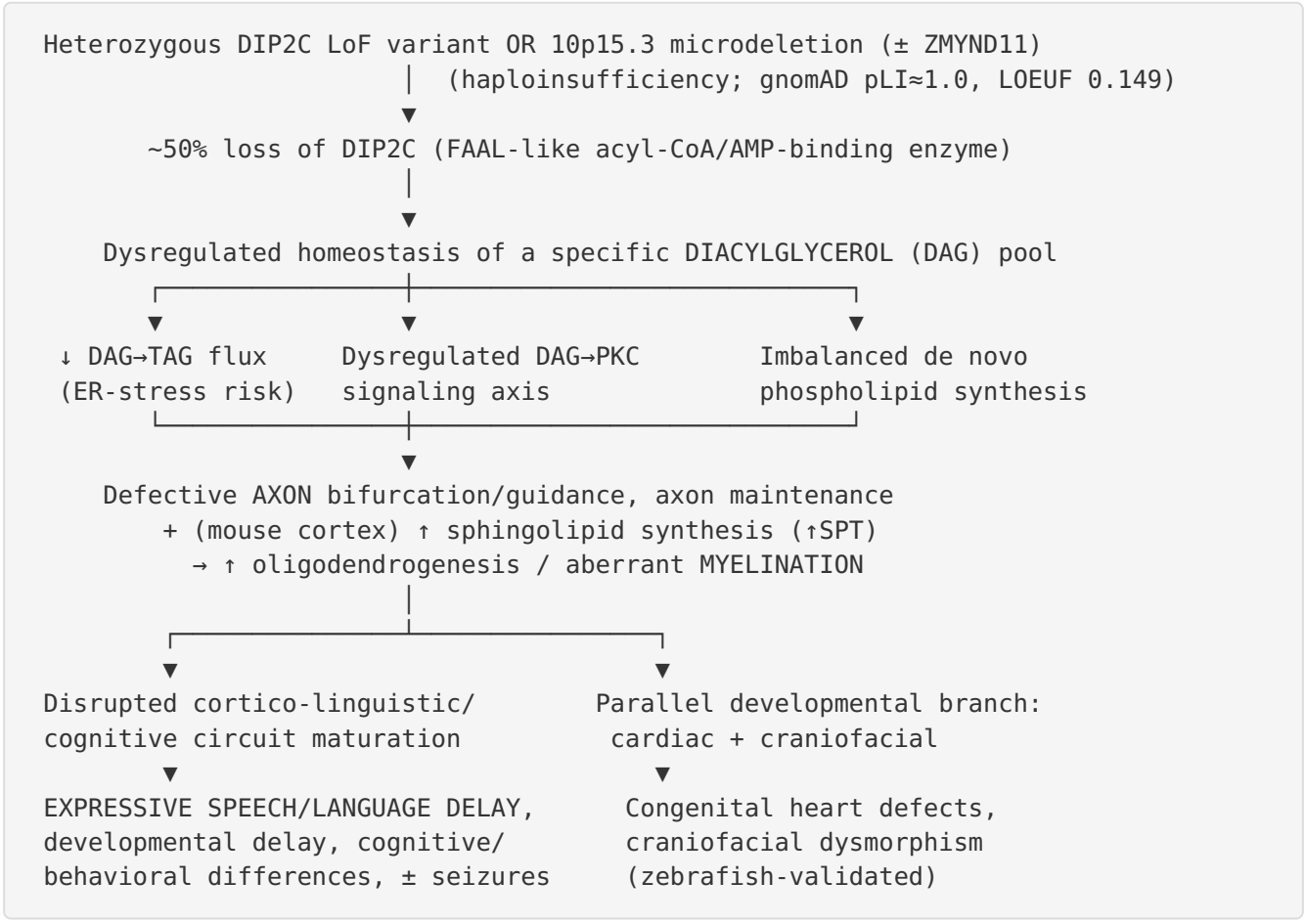
- **Taxonomy / orthologs:** *DIP2C* orthologs are broadly conserved and expressed in the developing nervous system from *Drosophila* to human. Orthologs: mouse *Dip2c* (mouse ortholog in MGI), zebrafish *dip2ca*, *C. elegans dip-2*, *Drosophila* DIP2, and yeast family members. DIP2 was studied as a conserved family *"using yeast, fly, and mouse models"* (PMID: 35766356).
- **Natural disease in other species:** no naturally occurring companion-animal or wildlife disease is catalogued (no OMIA entry noted); the animal evidence is from engineered models (below).
- **Comparative biology / conservation:** the neurodevelopmental axon-branching role and the DAG-lipid-homeostasis function are **evolutionarily conserved** across yeast, fly, worm, zebrafish, and mouse (PMID: 35766356, PMID: 27908785).
- **Transmission / zoonosis:** not applicable.

15. Model Organisms

Model	Type	Key phenotype / finding	Reference
Mouse (het & homozygous *Dip2c* mutants)	Mammalian, genetic	Homozygotes: weight loss, hyperlocomotion, cognitive impairment, abnormal lipid metabolism; heterozygotes: mild cognitive impairment — **recapitulating the dosage-sensitive human phenotype**; cortical sphingolipid-metabolism upregulation with elevated oligodendrogenesis/myelination	PMID: 41054236
Mouse (IMPC knockout)	Mammalian, genetic	11 significant genotype–phenotype associations; homozygotes: incomplete-penetrance preweaning lethality (MP:0011110), enlarged adrenal glands (MP:0000642), germ-cell defect, abnormal sternum; heterozygotes: small thymus, enlarged bladder/lymph nodes, abnormal uterus, increased monocytes (p=1.5e-6). **No behavioral/neurological deficit reached significance in the primary screen**	IMPC (Finding F007)
Mouse ES cells	Cellular, genetic	CRISPR/Cas9 *Dip2c* knockout to study early embryo development	PMID: 33813174
Zebrafish (*dip2ca* morpholino knockdown)	Vertebrate, induced	Decreased body length, craniofacial malformations, pericardial edema, linear heart tube, looping defects, arrhythmia; **rescued by wild-type *dip2ca*** — validates cardiac/craniofacial causality	PMID: 40915331
Drosophila (DIP2 mutant/RNAi)	Invertebrate, genetic	Ectopic axonal bifurcation and guidance defects in mushroom-body α/β neurons; downstream effector **Glaikit** (phospholipase D superfamily)	PMID: 27908785, PMID: 28465232
C. elegans (*dip-2*)	Invertebrate, genetic	Required for axon maintenance/regeneration; counterbalances de novo phospholipid (Kennedy pathway; CEPT-2/EPT-1) synthesis	PMID: 39974891
Yeast / fly / mouse (DIP2 family)	Cross-species	DIP2 as conserved homeostatic regulator of a DAG subset; FAAL-like domains redirect DAG→triacylglycerol; mitochondria–vacuole contact-site localization	PMID: 35766356

Phenotype recapitulation. The **Chai et al. mouse** is the strongest model of the human disorder, reproducing dosage-sensitive cognitive impairment and providing the sphingolipid/myelination mechanism: *"Homozygous mutant mice exhibited weight loss, hyperlocomotion, cognitive impairment, and abnormal lipid metabolism, whereas heterozygous mutant mice displayed only mild cognitive impairment, recapitulating the dosage-sensitive phenotype observed in human 10p15.3 microdeletion syndrome"* (PMID: 41054236). The **zebrafish** model validates the cardiac/craniofacial branch. Invertebrate models establish the conserved axon-development/lipid mechanism. **Limitations.** The IMPC primary screen did not detect speech/cognitive-relevant neurobehavioral deficits; no model captures human expressive-speech phenotypes directly (an intrinsic cross-species limitation).

Mechanistic Model / Interpretation



The unifying interpretation is that **DIP2C is a dosage-sensitive lipid-metabolic enzyme**; halving its dose destabilizes a narrow DAG/phospholipid/sphingolipid homeostatic set-point that neurons (and developing cardiac/craniofacial tissue) depend on for correct axon wiring and myelination. Because expressive-language circuits appear especially sensitive, the most penetrant clinical signal is speech delay, while the remaining features reflect variable expressivity of the same lipid-developmental perturbation. Upstream = the mutation and lipid dysregulation; downstream = axon/myelination defects and the clinical phenotype. The human→animal mechanistic bridge (step 7) remains **inferred** rather than directly demonstrated in human neurons.

Evidence Base

PMID	Title (abbrev.)	Role in this report
38421105	*De novo variants predicting haploinsufficiency for DIP2C are associated with expressive speech delay*	**Defining cohort** (23 individuals); phenotype, variant spectrum, inheritance, cardiac defects, protein domains
22847950	*Subtelomeric deletion of chromosome 10p15.3*	Establishes *DIP2C*/*ZMYND11* critical region; deletion-cohort phenotype frequencies; CMA diagnosis
34818214	*Intragenic deletion of ZMYND11 in 10p15.3*	Supports haploinsufficiency mechanism and shared deletion phenotype
39696561	*Prenatal diagnosis of pure 10p15.3 microdeletion*	Inheritance from a mildly affected parent → incomplete penetrance
35766356	*DIP2 is a unique regulator of diacylglycerol lipid homeostasis*	Core enzymatic mechanism (DAG→TAG, FAAL-like domains, mito-vacuole sites)
40327034	*Emergence of Dip2-mediated DAG-based PKC signalling*	DAG→PKC signaling axis
39974891	*C. elegans DIP-2 / phospholipid synthesis*	Axon maintenance/regeneration via phospholipid balance
27908785	*DIP2 regulates axonal bifurcation in Drosophila MB neurons*	Conserved axon-branching/guidance role; FAAL domains essential
28465232	*Glaikit downstream of DIP2*	Downstream effector of DIP2 axon pathway
41054236	*DIP2C deficiency leads to abnormal sphingolipid metabolism in mice*	Dosage-sensitive cognitive phenotype + sphingolipid/myelination mechanism
40915331	*DIP2C deficiency and congenital heart defects*	Zebrafish-validated cardiac/craniofacial causality
35987071	*DIP2C polymorphisms and ASD*	ASD susceptibility association
34617658	*Novel DIP2C splice variant in focal infantile epilepsy*	Epilepsy phenotype; variant c.1057+2T>G
33813174	*Knockout of Dip2c in murine ES cells*	ES-cell model resource
30950019	*10p15.3 microdeletion by WES*	Deletion-case phenotype; diagnostic modality

PMID	Title (abbrev.)	Role in this report
40049822	*r(10)/monosomy 10 mosaicism*	Structural-abnormality context; prenatal growth restriction
31542096	*Inv dup del(10p)*	Structural-rearrangement context

Beyond the core speech-delay phenotype, DIP2C is implicated across the neurodevelopmental continuum: *"Disco-interacting protein 2 C (DIP2C) has recently been reported as a new susceptibility gene for autism spectrum disorder (ASD) in a genome-wide association study"* (PMID: 35987071), and in focal infantile epilepsy where *"The de novo variation (NM_014974.3: c.1057+2T>G) in DIP2C was uncovered through whole-exome sequencing"* (PMID: 34617658).

Evidence-type mix: human clinical (cohorts/case reports), model organism (mouse, zebrafish, fly, worm, yeast), in vitro (minigene, ES cells), and computational (gnomAD constraint, Mendelian randomization). The human phenotype rests on **aggregated small cohorts**; the mechanism rests heavily on **model-organism and biochemical** data.

Limitations and Knowledge Gaps

1. **Small sample size.** The entire human evidence base is ~23 intragenic cases plus scattered deletion reports — insufficient for robust penetrance, genotype–phenotype, or prevalence estimates.
2. **Contiguous-gene confound.** In 10p15.3 deletions, *ZMYND11* co-deletion contaminates phenotype attribution; only intragenic *DIP2C* variants cleanly isolate the *DIP2C* contribution.
3. **Human→mechanism gap.** The causal link from lipid/axon/myelination defects (animal data) to human expressive-speech delay is **inferred**, not demonstrated in human neurons or patient-derived cells.
4. **No biomarker.** There is no biochemical or imaging biomarker; no validated episignature despite the DMAP1-binding domain.
5. **Incomplete model coverage.** The IMPC primary screen did not surface neurobehavioral/speech-relevant deficits; no model captures the human language phenotype.
6. **Unknown epidemiology.** Prevalence, incidence, sex ratio, and ethnic/geographic distribution are undefined.

7. **No therapeutics.** Zero registered trials; no gene-targeted or lipid-pathway-directed therapy evaluated.

Proposed Follow-up Experiments / Actions

1. **Expand the human cohort** via GeneMatcher/registries to refine penetrance, expressivity, genotype–phenotype correlation (LoF vs missense; intragenic vs deletion), and derive frequency estimates for each HPO feature.
2. **Patient-derived iPSC neurons/organoids** to test whether DIP2C haploinsufficiency perturbs DAG/sphingolipid pools, axon branching, and myelination in a human context — closing the human→mechanism gap.
3. **Targeted lipidomics** (DAG, TAG, phospholipid, sphingolipid species) in patient fibroblasts/plasma to search for a candidate biochemical biomarker and to test the serine-palmitoyltransferase/sphingolipid axis clinically.
4. **Conditional, brain-specific *Dip2c* mouse models** with language-relevant behavioral batteries (ultrasonic vocalization, learning) to better model the speech/cognitive phenotype absent from the IMPC screen.
5. **Systematic cardiac and EEG screening** across the cohort to quantify true frequencies of congenital heart defects and seizures and define surveillance guidelines.
6. **Mechanism-guided therapeutic exploration** — test whether modulating DAG→PKC signaling or sphingolipid biosynthesis (e.g., serine-palmitoyltransferase) rescues neurodevelopmental phenotypes in zebrafish/mouse models.
7. **Methylation/episignature study** leveraging the DMAP1-binding domain to determine whether *DIP2C* LoF produces a reproducible epigenetic classifier usable diagnostically.

Report compiled from 10 confirmed findings and 17 reviewed papers across 5 investigation iterations. Evidence sources span human clinical cohorts, model organisms (mouse, zebrafish, Drosophila, C. elegans, yeast), in vitro systems, and computational constraint analysis.
